

Physical Chemistry (II)

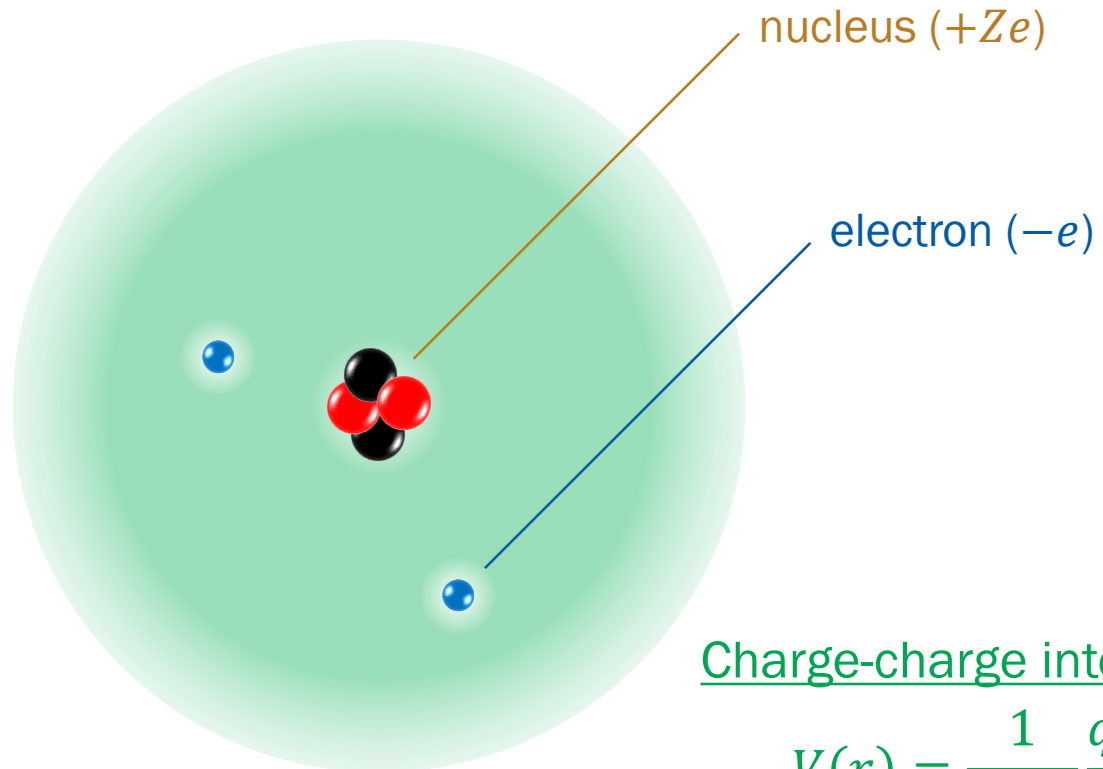
Lecture 10

Hydrogen Atom



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Atom

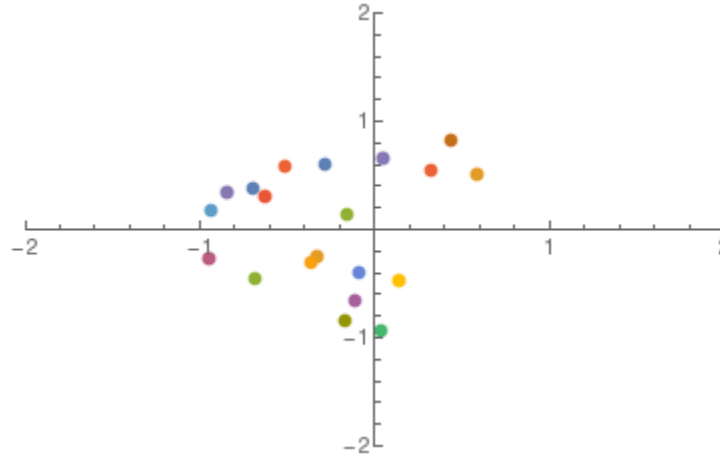


Charge-charge interaction

$$V(r) = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r}$$

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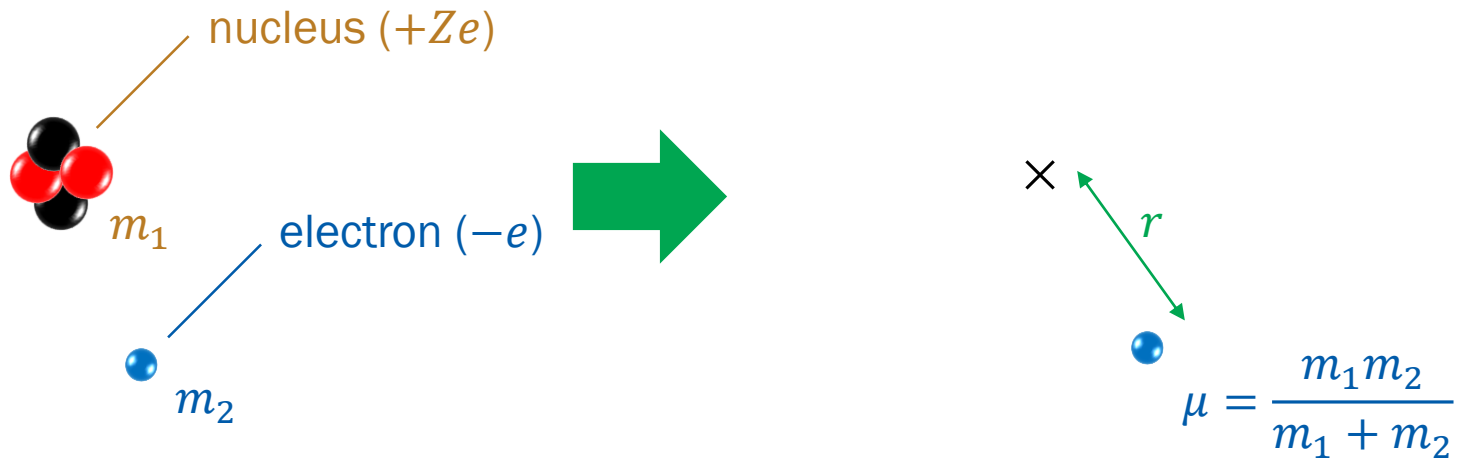
N-Body Problem ($N \geq 3$)



Mathematically impossible to solve exactly
even when we know the exact form of potential

Hydrogen-like Atom

one nucleus + one electron



$$V(r) = -\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r}$$

Quantum Mechanics

$$\left[-\frac{\hbar^2}{2\mu} \nabla^2 + \mathcal{V}(\hat{x}, \hat{y}, \hat{z}) \right] \psi(\vec{r}) = E\psi(\vec{r})$$

$$\left[-\frac{\hbar^2}{2\mu} \nabla^2 + \mathcal{V}(\hat{r}) \right] \psi(r, \theta, \phi) = E\psi(r, \theta, \phi)$$

In the spherical coordinates,

$$\begin{aligned} -\frac{\hbar^2}{2\mu} \left[\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \psi(r, \theta, \phi) \\ + \mathcal{V}(\hat{r}) \psi(r, \theta, \phi) = E\psi(r, \theta, \phi) \end{aligned}$$

Separation of Variables

Assume $\psi(r, \theta, \phi) = R(r)Y(\theta, \phi)$;

$$\begin{aligned} & -\frac{1}{R(r)} \left(r^2 \frac{\partial^2}{\partial r^2} + 2r \frac{\partial}{\partial r} \right) R(r) + \frac{2\mu}{\hbar^2} r^2 [\mathcal{V}(\hat{r}) - E] \\ & = \frac{1}{Y(\theta, \phi)} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] Y(\theta, \phi) \\ & = \lambda \text{ (constant)} \end{aligned}$$

Two Equations

$$-\frac{\hbar^2}{2\mu} \left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} + \frac{\lambda}{r^2} \right) R(r) + \mathcal{V}(\hat{r})R(r) = ER(r)$$

$$\left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] Y(\theta, \phi) = \lambda Y(\theta, \phi)$$

We know the solution of the second equation!

Hint from Rigid Rotor

$$\left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] Y(\theta, \phi) = \lambda Y(\theta, \phi)$$

Recall the Schrödinger equation for a rigid rotator:

$$-\frac{\hbar^2}{2I} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \psi(\theta, \phi) = E \psi(\theta, \phi)$$

whose energy $E = \frac{\hbar^2}{2I} J(J + 1)$ and wavefunction $\psi = Y_J^m$

Hence, $\lambda = -l(l + 1)$ and $Y(\theta, \phi) = Y_l^{m_l}(\theta, \phi)$

Radial Equation

$$-\frac{\hbar^2}{2\mu} \left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} - \frac{l(l+1)}{r^2} \right) R(r) + \mathcal{V}(\hat{r}) R(r) = ER(r)$$

$$\mathcal{V}(\hat{r}) = -\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{\hat{r}} = -\frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r}$$

Difficult to solve, but (again) we know the solution

Solution for Radial Equation

$$R_{nl}(r) = \left\{ \left(\frac{2Z}{\underbrace{na_0}_{\substack{\text{Bohr radius} \\ 4\pi\epsilon_0\hbar^2/\mu e^2}}} \right)^3 \frac{(n-l-1)!}{2n[(n+l)!]^3} \right\}^{1/2} \left(\frac{2Zr}{na_0} \right)^l e^{-Zr/na_0} \underbrace{L_{n+l}^{2l+1} \left(\frac{2Zr}{na_0} \right)}_{\substack{\text{associated Laguerre} \\ \text{polynomial}}}$$

A few associated Laguerre polynomials:

$$L_1^1(x) = 1$$

$$L_2^1(x) = -2x + 4$$

$$L_3^1(x) = 3x^2 - 18x + 18$$

⋮

Solution for Schrödinger Equation

- Quantum numbers: n, l, m_l , such that

$$n = 1, 2, 3, \dots; \quad * K, L, M, \dots$$

$$l = 0, 1, 2, \dots, n - 1; \quad * s, p, d, f, g, \dots$$

$$m_l = 0, \pm 1, \pm 2, \dots, \pm l$$

- Wavefunction (a.k.a. “orbital”):

$$\psi_{nlm_l}(r, \theta, \phi) = R_{nl}(r)Y_l^{m_l}(\theta, \phi)$$

Solution for Schrödinger Equation

- Quantum numbers: n, l, m_l , such that

$$n = 1, 2, 3, \dots;$$

$$l = 0, 1, 2, \dots, n - 1;$$

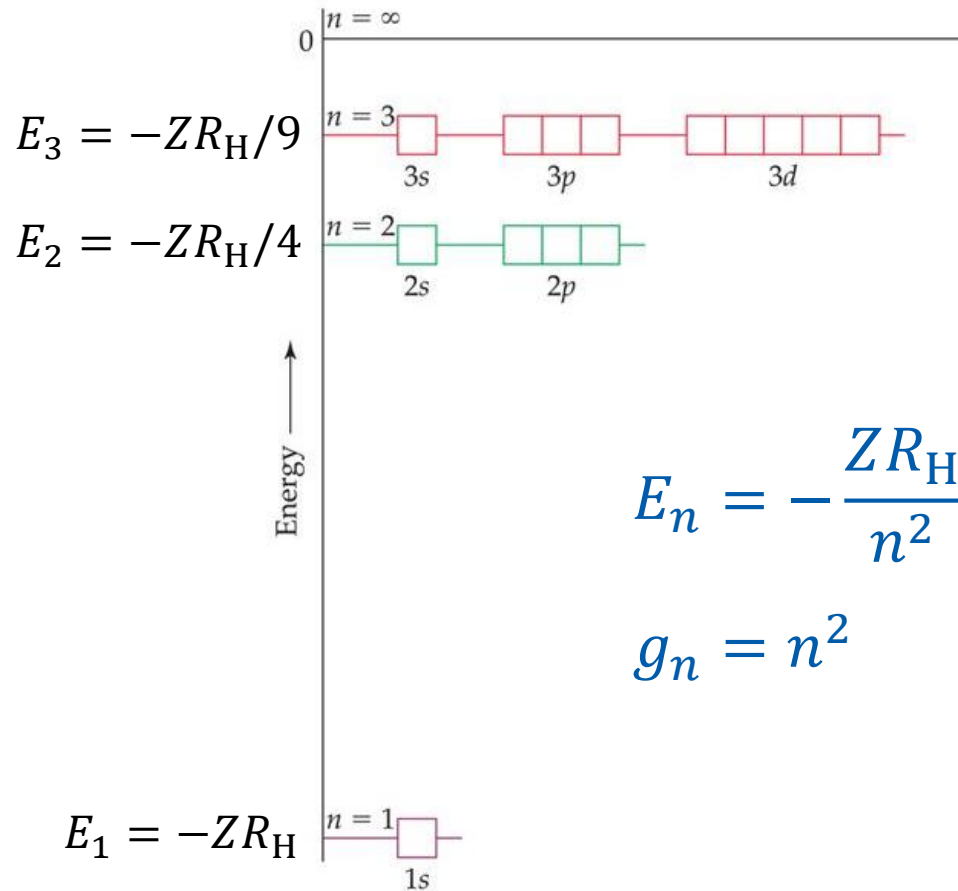
$$m_l = 0, \pm 1, \pm 2, \dots, \pm l$$

- Energy:

$$E_{nlm_l} = -\frac{Ze^2}{8\pi\epsilon_0 a_0} \frac{1}{n^2} = -\frac{ZR_H}{n^2} = E_n$$

RS 6.3
M 7.1

Energy Levels



(Image: <https://slideplayer.com/slide/5959399/>)

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Populations for Hydrogen Atom

$$P(E) \propto e^{-E/k_{\text{B}}T}$$

$$E_1 = -R_{\text{H}} \text{ and } E_2 = -R_{\text{H}}/4$$

At 298 K,

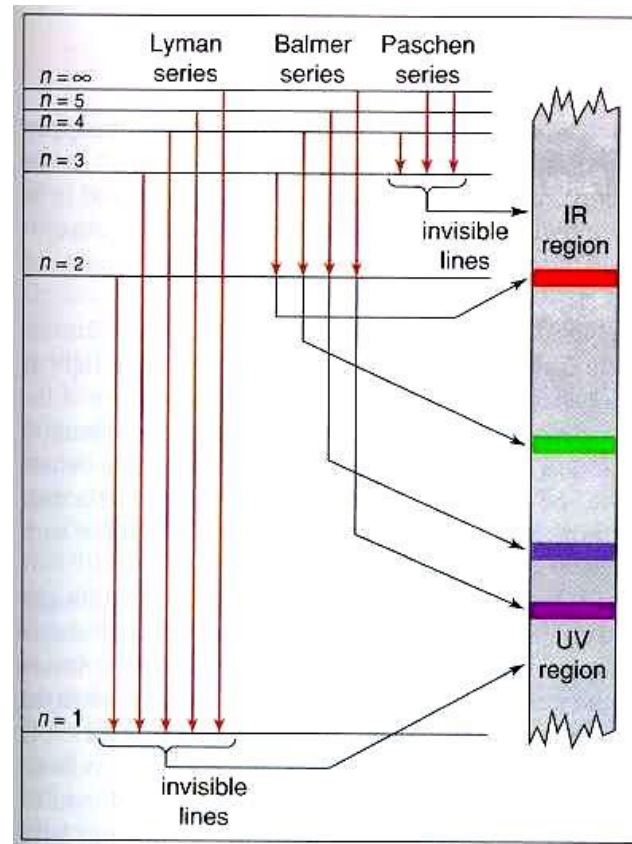
$$P(E_1) \approx 1 \text{ and } P(E_2) \approx e^{-3R_{\text{H}}/4k_{\text{B}}T} = e^{-530} \approx 10^{-231}$$

At room temperature, hydrogen atoms are basically
in the ground electronic state



Spectroscopy

- No selection rule for n
- This is why the Bohr atom was successful



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Wavefunctions: Eigenfunctions

$$\psi_{nlm_l}(r, \theta, \phi) = R_{nl}(r)Y_l^{m_l}(\theta, \phi)$$

$$\hat{H}\psi_{nlm_l} = E_n\psi_{nlm_l}$$

$$\hat{L}^2 Y_l^{m_l} = l(l+1)\hbar^2 Y_l^{m_l} \rightarrow \hat{L}^2 \psi_{nlm_l} = l(l+1)\hbar^2 \psi_{nlm_l}$$

$$\hat{L}_z Y_l^{m_l} = m_l \hbar Y_l^{m_l} \rightarrow \hat{L}_z \psi_{nlm_l} = m_l \hbar \psi_{nlm_l}$$

A Few Wavefunctions

With $\rho = Zr/a_0$,

$$\psi_{100} = \frac{1}{\sqrt{\pi}} \left(\frac{Z}{a_0} \right)^{3/2} e^{-\rho}$$

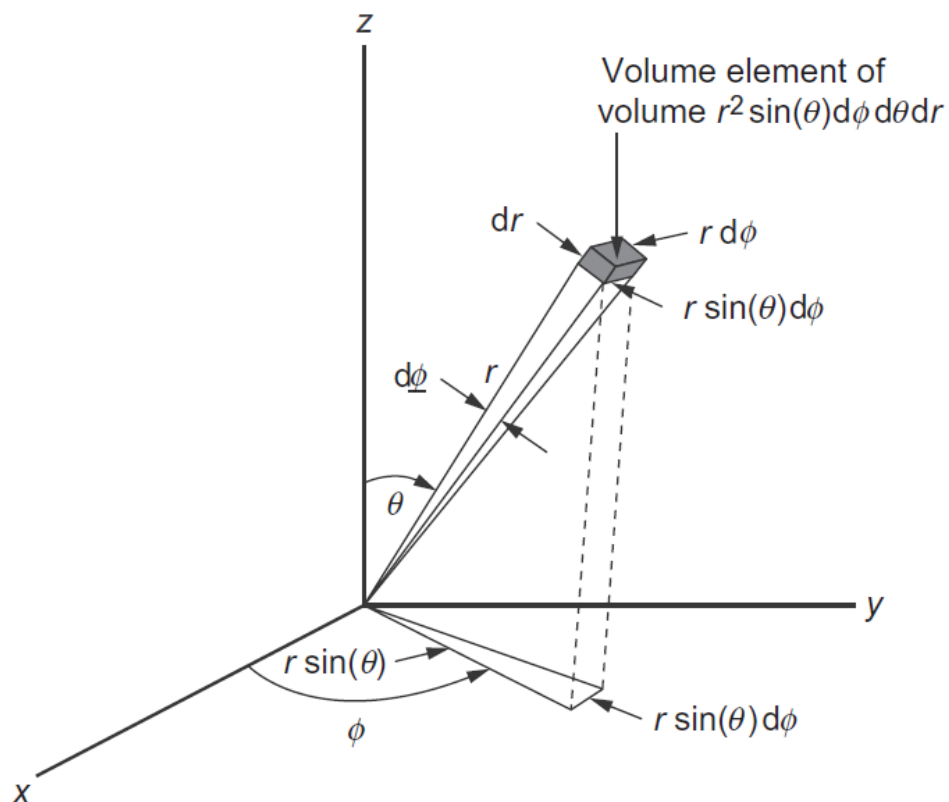
$$\psi_{200} = \frac{1}{4\sqrt{2\pi}} \left(\frac{Z}{a_0} \right)^{3/2} (2 - \rho) e^{-\rho/2}$$

$$\psi_{210} = \frac{1}{4\sqrt{2\pi}} \left(\frac{Z}{a_0} \right)^{3/2} \rho e^{-\rho/2} \cos \theta$$

$$\psi_{21\pm 1} = \mp \frac{1}{8\sqrt{\pi}} \left(\frac{Z}{a_0} \right)^{3/2} \rho e^{-\rho/2} \sin \theta e^{\pm i\phi}$$

⋮

Integration in Spherical Coord.



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 9.5)

Normalization and Orthogonality

- Normalization condition

$$\int_0^{2\pi} \int_0^{\pi} \int_0^{\infty} \psi^*(r, \theta, \phi) \psi(r, \theta, \phi) r^2 \sin \theta \, dr \, d\theta \, d\phi = 1$$

- Orthogonality condition ($a \neq b$)

$$\int_0^{2\pi} \int_0^{\pi} \int_0^{\infty} \psi_a^*(r, \theta, \phi) \psi_b(r, \theta, \phi) r^2 \sin \theta \, dr \, d\theta \, d\phi = 0$$

Example 7-1

Show that ψ_{210} is normalized and that it is orthogonal to ψ_{200} .

$$\psi_{210}(r, \theta, \phi) = \frac{1}{4\sqrt{2\pi}} \left(\frac{Z}{a_0}\right)^{5/2} r e^{-Zr/2a_0} \cos \theta$$

The normalization condition is

$$\int_0^{2\pi} \int_0^\pi \int_0^\infty \psi_{210}^*(r, \theta, \phi) \psi_{210}(r, \theta, \phi) r^2 \sin \theta \, dr \, d\theta \, d\phi = 1$$

The left-hand side is

$$\text{LHS} = \frac{1}{32\pi} \left(\frac{Z}{a_0}\right)^5 \int_0^{2\pi} \int_0^\pi \int_0^\infty e^{-Zr/a_0} r^2 \cos^2 \theta r^2 \sin \theta \, dr \, d\theta \, d\phi$$

Example 7-1

Show that ψ_{210} is normalized and that it is orthogonal to ψ_{200} .

$$\begin{aligned}\text{LHS} &= \frac{1}{32\pi} \left(\frac{Z}{a_0} \right)^5 \int_0^{2\pi} \int_0^\pi \int_0^\infty r^4 e^{-Zr/a_0} \cos^2 \theta \sin \theta \, dr \, d\theta \, d\phi \\ &= \frac{1}{32\pi} \left(\frac{Z}{a_0} \right)^5 \int_0^\infty r^4 e^{-Zr/a_0} \, dr \int_0^\pi \cos^2 \theta \sin \theta \, d\theta \int_0^{2\pi} d\phi \\ &= \frac{1}{32\pi} \times 24 \times \frac{2}{3} \times 2\pi = 1\end{aligned}$$

Hence, ψ_{210} is normalized.

Example 7-1

Show that ψ_{210} is normalized and that it is orthogonal to ψ_{200} .

$$\psi_{210}(r, \theta, \phi) = \frac{1}{4\sqrt{2\pi}} \left(\frac{Z}{a_0}\right)^{5/2} r e^{-Zr/2a_0} \cos \theta$$

$$\psi_{200}(r, \theta, \phi) = \frac{1}{4\sqrt{2\pi}} \left(\frac{Z}{a_0}\right)^{3/2} \left(2 - \frac{Zr}{a_0}\right) e^{-Zr/2a_0}$$

The orthogonality condition is

$$\int_0^{2\pi} \int_0^\pi \int_0^\infty \psi_{210}^*(r, \theta, \phi) \psi_{200}(r, \theta, \phi) r^2 \sin \theta \, dr \, d\theta \, d\phi = 0$$

Example 7-1

Show that ψ_{210} is normalized and that it is orthogonal to ψ_{200} .

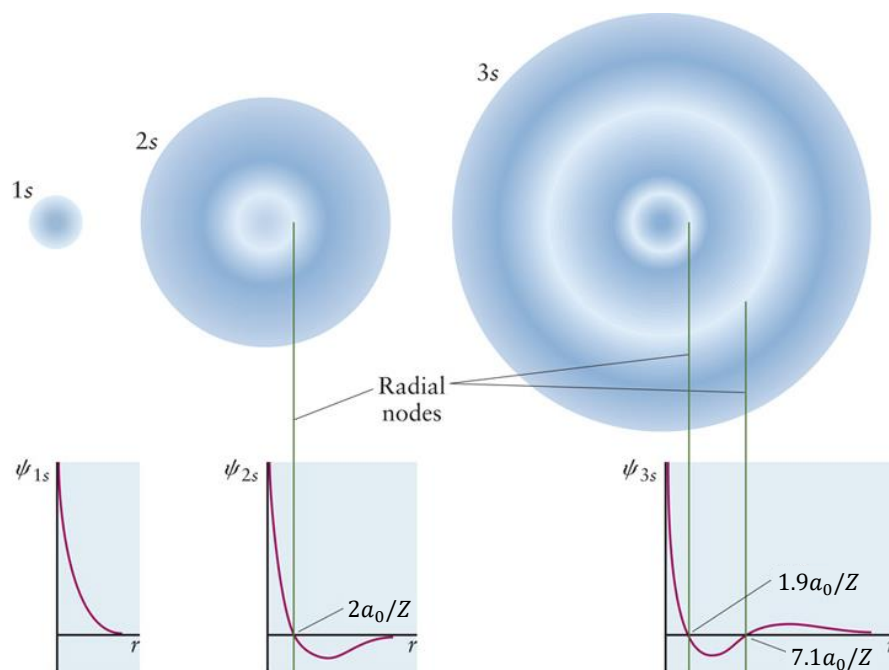
$$\begin{aligned}\text{LHS} &= \frac{1}{32\pi} \left(\frac{Z}{a_0} \right)^4 \int_0^{2\pi} \int_0^\pi \int_0^\infty r^3 \left(2 - \frac{Zr}{a_0} \right) e^{-Zr/a_0} \cos \theta \sin \theta \, dr \, d\theta \, d\phi \\ &= \frac{1}{32\pi} \left(\frac{Z}{a_0} \right)^4 \int_0^\infty r^3 \left(2 - \frac{Zr}{a_0} \right) e^{-Zr/a_0} \, dr \int_0^\pi \cos \theta \sin \theta \, d\theta \int_0^{2\pi} d\phi\end{aligned}$$

The θ integral vanishes, and ψ_{210} is orthogonal to ψ_{200} .

RS 6.4
M 7.2

s Orbitals

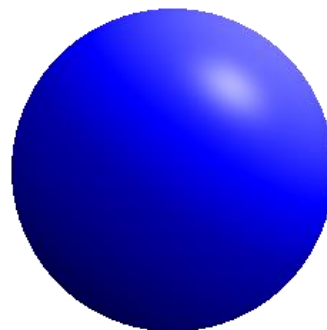
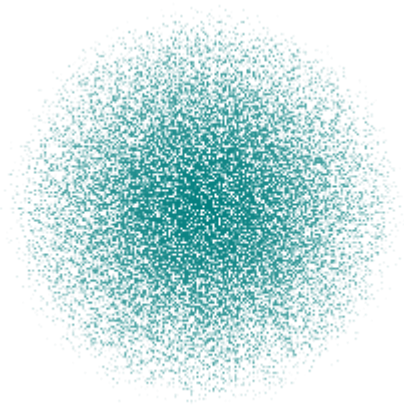
$$\psi_{n00}(r, \theta, \phi) = R_{n0}(r)Y_0^0(\theta, \phi) = \frac{1}{\sqrt{4\pi}} R_{n0}(r)$$



(Image: Oxtoby et al., *Principles of Modern Chemistry*, 7th Ed., Fig. 5.7)

RS 6.4
M 7.2

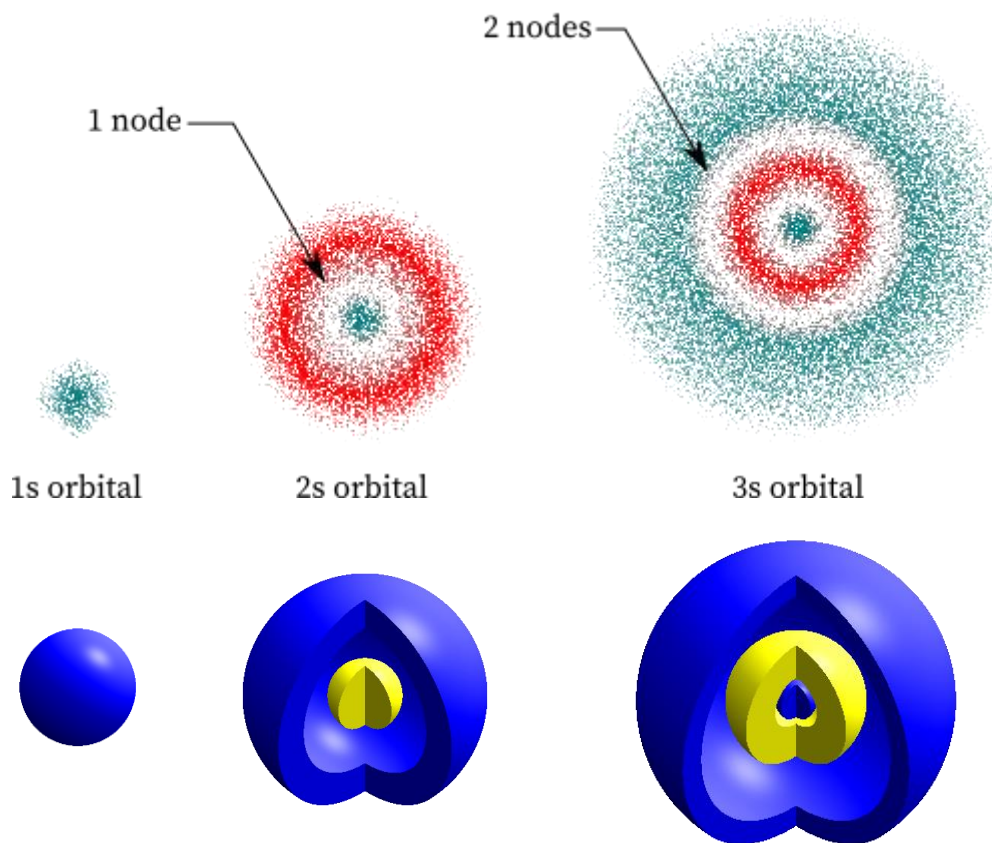
ψ_{100} : 1s Orbital



(Images: <https://chemistrygod.com/atomic-orbital>
https://commons.wikimedia.org/wiki/File:Atomic_orbitals_n123_m-eigenstates.png)

RS 6.4
M 7.2

ψ_{200} (2s) and ψ_{300} (3s)



p Orbitals

$$\psi_{n10}(r, \theta, \phi) = R_{n1}(r)Y_1^0(\theta, \phi) = \left(\frac{3}{4\pi}\right)^{1/2} R_{n1}(r) \cos \theta$$

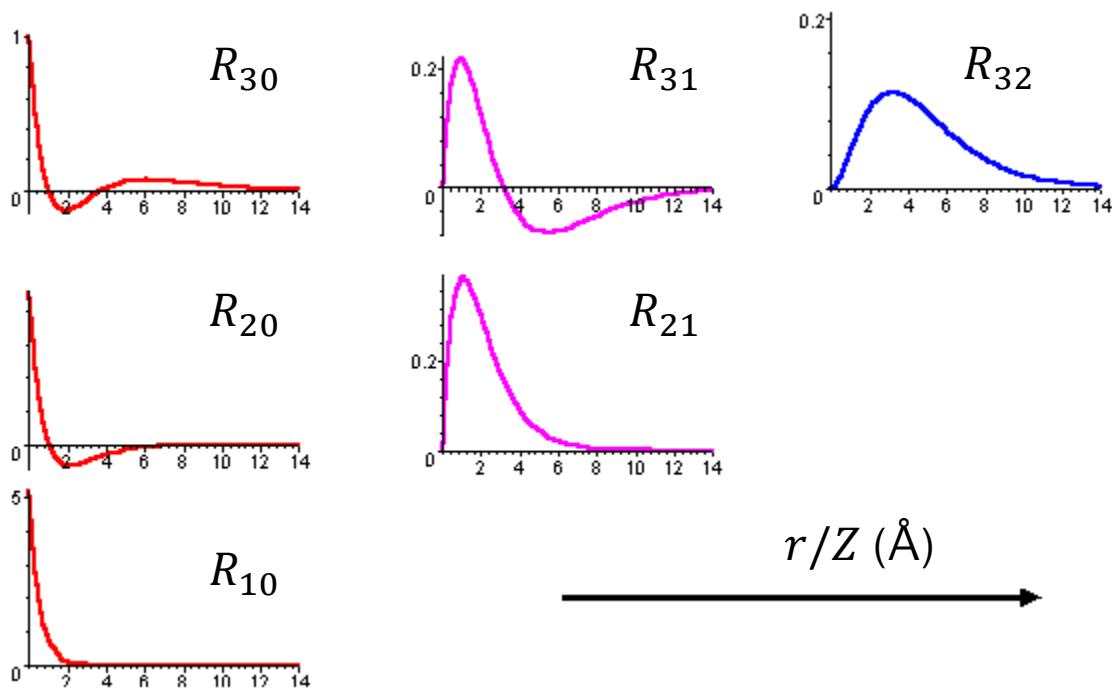
$$\psi_{n11}(r, \theta, \phi) = R_{n1}(r)Y_1^1(\theta, \phi) = -\left(\frac{3}{8\pi}\right)^{1/2} R_{n1}(r) \sin \theta e^{i\phi}$$

$$\psi_{n1-1}(r, \theta, \phi) = R_{n1}(r)Y_1^{-1}(\theta, \phi) = \left(\frac{3}{8\pi}\right)^{1/2} R_{n1}(r) \sin \theta e^{-i\phi}$$

$\psi_{n10}(r, \theta, \phi)$: real / $\psi_{n1\pm 1}(r, \theta, \phi)$: complex

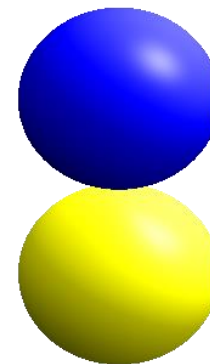
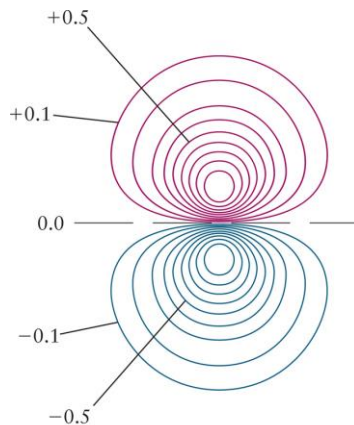
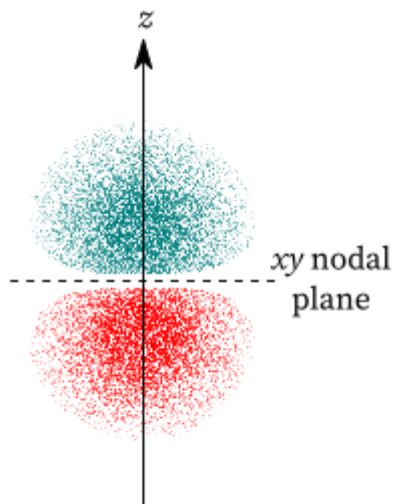
RS 6.4
M 7.3

Radial Parts R_{nl}



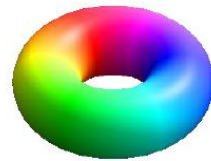
RS 6.4
M 7.3

ψ_{210} : $2p_0 = 2p_z$ Orbital

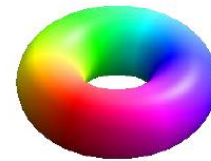


ψ_{211} ($2p_1$) and ψ_{21-1} ($2p_{-1}$)

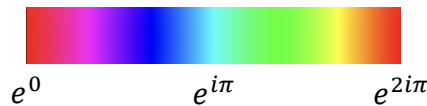
$$\psi_{21\pm 1} = \mp \frac{1}{8\sqrt{\pi}} \left(\frac{Z}{a_0} \right)^{3/2} \rho e^{-\rho/2} \sin \theta e^{\pm i\phi}$$



$2p_1$



$2p_{-1}$



(Image: https://commons.wikimedia.org/wiki/File:Atomic_orbitals_n123_m-eigenstates.png)

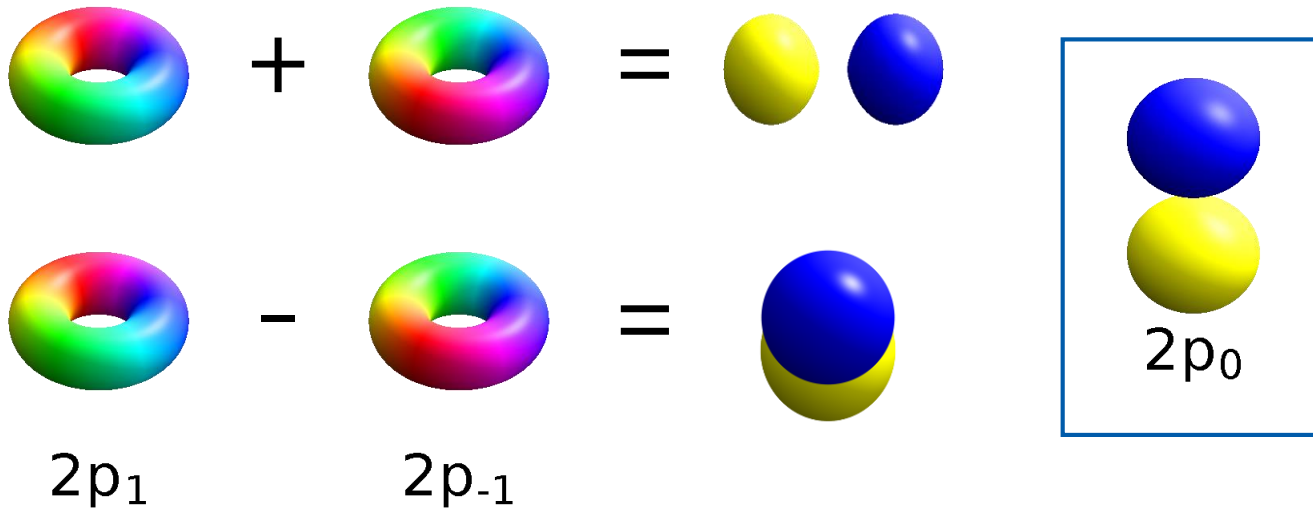
How to Obtain $2p_x$ and $2p_y$

$$\psi_{21\pm 1} = \mp \frac{1}{8\sqrt{\pi}} \left(\frac{Z}{a_0} \right)^{3/2} \rho e^{-\rho/2} \sin \theta e^{\pm i\phi}$$

$$\psi_{2p_x} = \frac{1}{\sqrt{2}} (-\psi_{211} + \psi_{21-1}) = \frac{1}{4\sqrt{2\pi}} \left(\frac{Z}{a_0} \right)^{3/2} \rho e^{-\rho/2} \sin \theta \cos \phi$$

$$\psi_{2p_y} = \frac{i}{\sqrt{2}} (\psi_{211} + \psi_{21-1}) = \frac{1}{4\sqrt{2\pi}} \left(\frac{Z}{a_0} \right)^{3/2} \rho e^{-\rho/2} \sin \theta \sin \phi$$

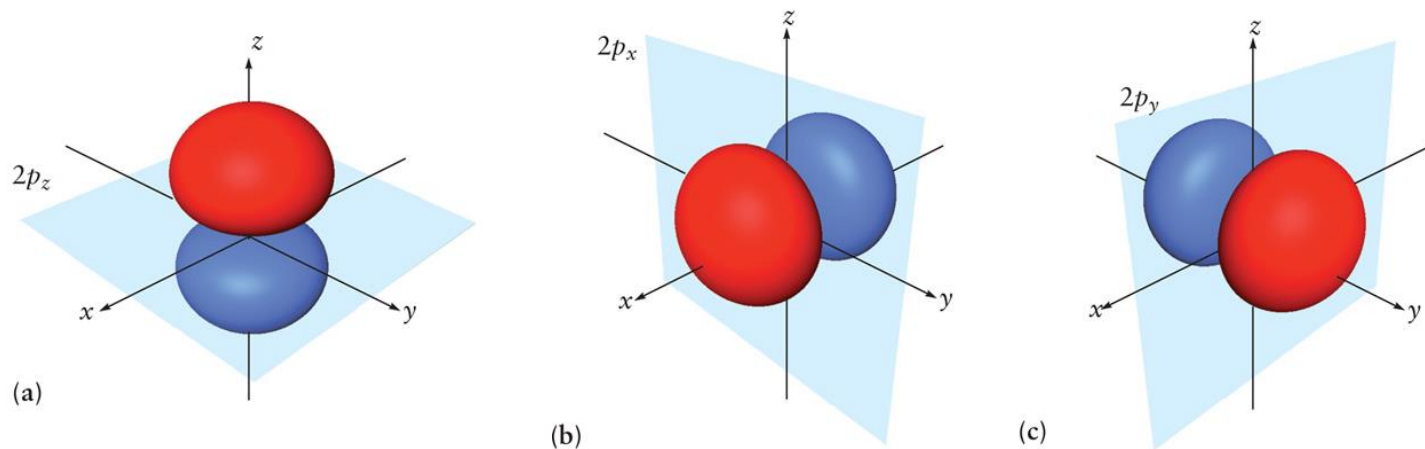
Mix and Match



(Image: https://commons.wikimedia.org/wiki/File:Atomic_orbitals_n123_m-eigenstates.png)

RS 6.4
M 7.3

Our Old Friends



Degeneracy Makes It All

- $2p_0, 2p_1, 2p_{-1}$ are all **degenerate** ($E_2 = -ZR_H/4$)
- Their linear combinations also satisfy the Schrödinger equation

$$\hat{H}\psi_{2p_z} = \hat{H}\psi_{210} = E_2\psi_{210}$$

$$\hat{H}\psi_{2p_x} = \hat{H}(\psi_{21-1} - \psi_{211})/\sqrt{2} = E_2(\psi_{21-1} - \psi_{211})/\sqrt{2} = E_2\psi_{2p_x}$$

$$\hat{H}\psi_{2p_y} = \hat{H}i(\psi_{21-1} + \psi_{211})/\sqrt{2} = E_2i(\psi_{21-1} + \psi_{211})/\sqrt{2} = E_2\psi_{2p_y}$$

Not in text

Angular Momenta?

$$\hat{L}^2 \psi_{nlm_l} = l(l+1)\hbar^2 \psi_{nlm_l}$$

$$\hat{L}_z \psi_{nlm_l} = m_l \hbar \psi_{nlm_l}$$

$$\hat{L}^2 \psi_{2p_x} = \hat{L}^2 (\psi_{21-1} - \psi_{211}) / \sqrt{2}$$

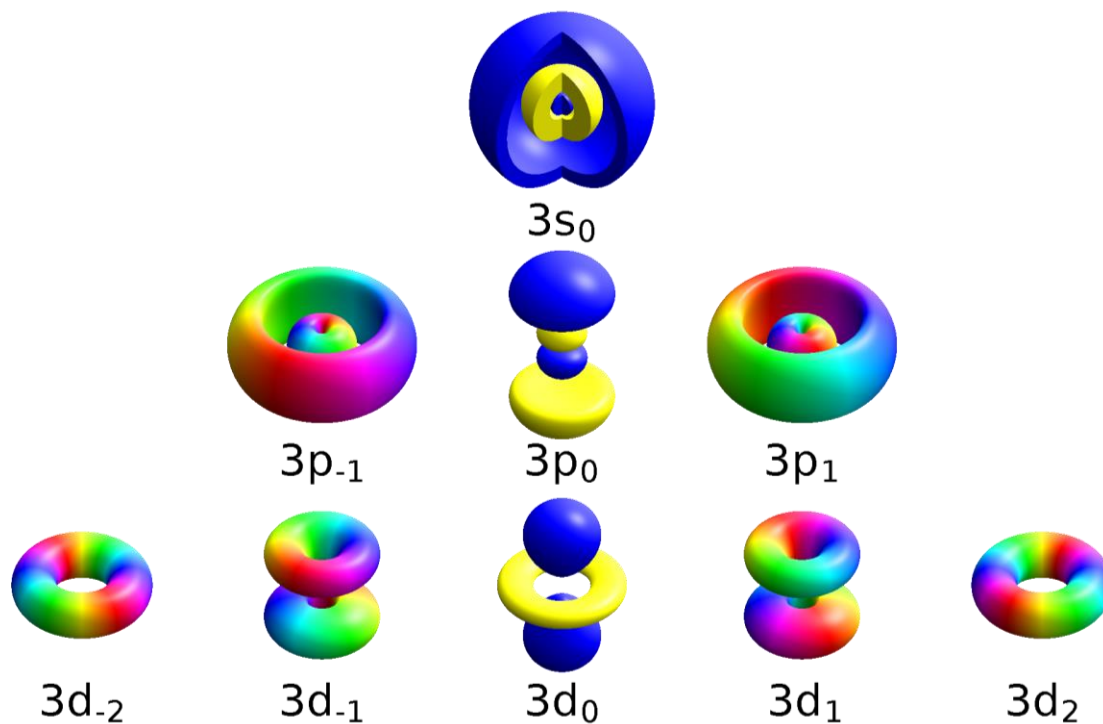
$$= l(l+1)\hbar^2 (\psi_{21-1} - \psi_{211}) / \sqrt{2} = l(l+1)\hbar^2 \psi_{2p_x}$$

$$\hat{L}_z \psi_{2p_x} = \hat{L}_z (\psi_{21-1} - \psi_{211}) / \sqrt{2}$$

$$= (-\hbar \psi_{21-1} - \hbar \psi_{211}) / \sqrt{2} \neq a \psi_{2p_x}$$

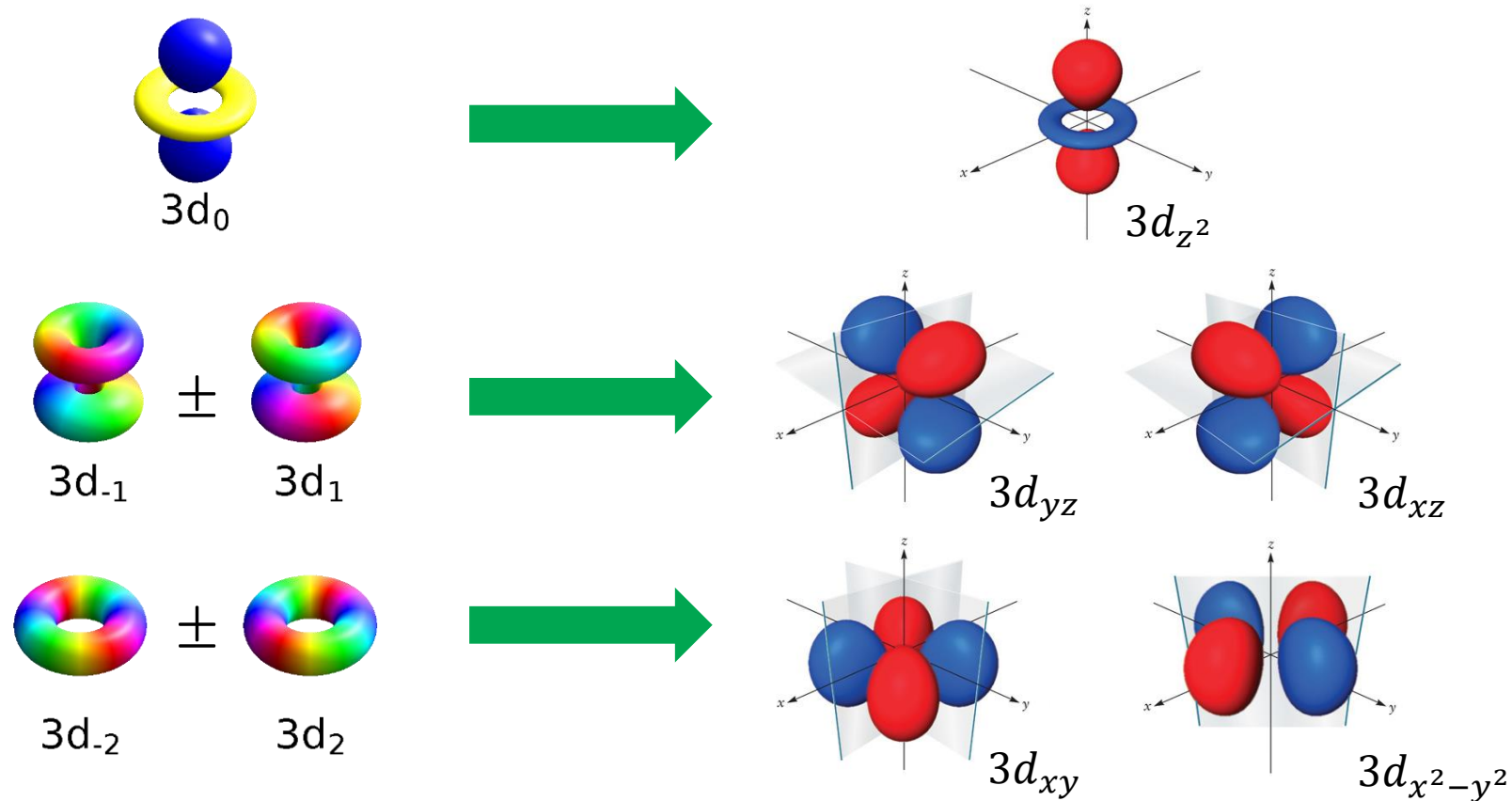
RS 6.4
M 7.3

$n = 3$ Orbitals



(Image: https://commons.wikimedia.org/wiki/File:Atomic_orbitals_n123_m-eigenstates.png)

Mix and Match: *d* Orbitals



(Images: https://commons.wikimedia.org/wiki/File:Atomic_orbitals_n123_m-eigenstates.png
Oxtoby et al., *Principles of Modern Chemistry*, 7th Ed., Fig. 5.13)